

Spin-Polarized Josephson Supercurrent in Nodeless Altermagnets

Chuang Li^{1,2,*} Jin-Xing Hou^{1,2,3,*} Fu-Chun Zhang,⁴ Song-Bo Zhang^{1,2,3,†} and Lun-Hui Hu^{1,‡}

¹Center for Correlated Matter and School of Physics, *Zhejiang University*, Hangzhou 310058, China

²Hefei National Laboratory, Hefei, Anhui 230088, China

³International Center for Quantum Design of Functional Materials (ICQD), *University of Science and Technology of China*, Hefei, Anhui 230026, China

⁴Kavli Institute for Theoretical Sciences, *University of Chinese Academy of Sciences*, Beijing 100190, China



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Long-range propagation of equal-spin triplet Cooper pairs typically occurs in ferromagnet-*s*-wave superconductor junctions, where net magnetization plays a crucial role. Here, we propose a fundamentally different scenario in which Josephson supercurrents mediated exclusively by spin-triplet pairings emerge in systems with *zero* net magnetization. We identify collinear altermagnets, particularly a subclass termed nodeless altermagnets, as ideal platforms to realize this phenomenon. These materials host spin-split Fermi surfaces that do not intersect altermagnetic nodal lines and support maximal spin-valley polarization, yielding fully spin-polarized electronic states at each valley. Consequently, Josephson junctions based on nodeless altermagnets sustain supercurrents solely through spin-polarized triplet pairing correlations, simultaneously contributed by spin-up Cooper pairs from one valley and spin-down Cooper pairs from the other. Furthermore, controlling the relative local inversion-symmetry breaking at the two interfaces enables a robust $0 - \pi$ transition without fine tuning, while adjusting the junction orientation allows a crossover between pure triplet and mixed singlet-triplet states. Our Letter thus establishes nodeless altermagnets as a unique platform for altermagnetic superconductors with magnetization-free spin-polarized supercurrents.

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Introduction—The recent discovery of collinear altermagnetism (AM) has significantly expanded our understanding of magnetic materials [1–12]. Unlike conventional antiferromagnets, AM hosts antiparallel spins coupled through crystalline symmetries such as rotation and reflection, establishing a new magnetic phase characterized by vanishing net magnetization and momentum-dependent spin splitting [13–16]. This unconventional magnetic phase can be realized in diverse systems [17–30], and manifests a range of novel quantum phenomena including nonrelativistic spin splitting [10], crystal-symmetry-paired spin-valley locking (SVL) [8,31], spin-orbital textures [32,33], and anomalous transport properties [34–41]. Recent experiments have observed both spin-splitting and SVL in various quantum materials [11,42–52]. While momentum-space spin splitting may also arise from mechanisms like spin-channel Pomeranchuk instabilities [53–55] or *d*-wave spin-density wave states [56], SVL is unique to AMs thus far.

SVL represents a distinctive manifestation of spin splitting under specific symmetry constraints [31]. When spin-orbit coupling is negligible, the spin-space group forbids spin splitting along certain momentum directions.

For example, the coexistence of symmetries $[C_2||\mathcal{M}_{[11]}]$ and $[C_2||\mathcal{M}_{[\bar{1}\bar{1}]}]$ guarantee vanishing spin-splitting along the $k_x = \pm k_y$ directions, resulting in symmetry-protected altermagnetic nodal lines. Depending on whether these nodal lines intersect the Fermi surface, AMs can be divided into two classes: *nodal* AMs and *nodeless* AMs [13]. This classification distinguishes different Fermi surface topologies and is relevant only for metallic phases. Particularly, nodeless AMs feature spin-split Fermi surfaces that avoid enclosing the Γ point and inherently support SVL. Hence, SVL is a defining characteristic of nodeless AMs. While nodal AMs have been well explored [57–68], nodeless AMs present fundamentally distinct opportunities. In particular, SVL in AMs breaks time-reversal symmetry, a feature that remains underexplored but with great potential for spintronics applications and superconducting proximity effects.

In this Letter, we demonstrate nodeless AM as ideal platforms for proximity-induced *pure* spin-triplet correlations without invoking net magnetization. For AM-based Josephson junctions, Fig. 1(a) shows that nodal AMs permit spatially oscillating spin-singlet pairing [57,58]. In contrast, nodeless AMs uniquely generate spin-polarized triplet correlations containing both $c_{\uparrow}c_{\uparrow}$ (from *X* valley) and $c_{\downarrow}c_{\downarrow}$ (from *Y* valley) [Fig. 1(b)]. This valley degree of freedom facilitates an experimentally tunable, orientation-dependent Josephson effect, allowing controlled crossover

*These authors contributed equally to this work.

†Contact author: songbozhang@ustc.edu.cn

‡Contact author: lunhui@zju.edu.cn

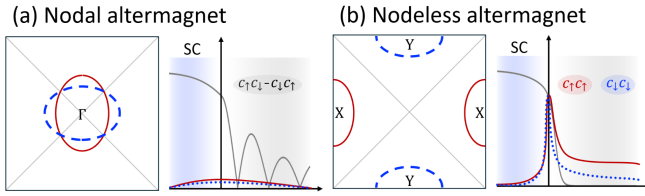


FIG. 1. Proximity effects in s -wave superconductor/AM junctions, showing dominant pairing correlations: (a) Spin-singlet pairing with spatial oscillations in nodal AM metals [57,58]. (b) Spin-triplet pairing in nodeless AM metals, with $c_{\uparrow}c_{\uparrow}$ and $c_{\downarrow}c_{\downarrow}$ contributed from two valleys, respectively. Left panels: Fermi surfaces with solid and dashed lines indicating spin-up and spin-down polarizations. Right panels: corresponding pairing correlations across the junction.

from pure triplet to mixed singlet-triplet states. Tuning the relative local inversion-symmetry breaking at the two junction interfaces triggers a robust $0 - \pi$ transition without fine tuning. Our Letter provides an extrinsic mechanism yielding exotic altermagnetic superconductors with spin-polarized supercurrents that break spin-space group symmetries [69].

Spin-valley polarization in AMs—To quantify SVL in AMs, we introduce the spin-valley polarization ΔP . Consider a system with two inequivalent valleys, X and Y . The spin polarization around each valley is $P_{X(Y)} = (\mathcal{N}_{X(Y),\uparrow} - \mathcal{N}_{X(Y),\downarrow}) / (\mathcal{N}_{X(Y),\uparrow} + \mathcal{N}_{X(Y),\downarrow})$, where $\mathcal{N}_{X(Y),\sigma}$ denotes the spin-resolved density of states at valley X or Y for spin $\sigma \in \{\uparrow, \downarrow\}$. The spin-valley polarization

is then defined by $\Delta P \equiv P_X P_Y$. In normal metals with spin-degenerate bands, the spin polarization vanishes, resulting in $\Delta P = 0$. In contrast, AMs feature vanishing net magnetization, which holds $P_X + P_Y = 0$, leading to $\Delta P < 0$. While ΔP can take any value in between -1 and 0 , it becomes quantized to -1 when each valley hosts fully spin-polarized Fermi surfaces. This represents perfect SVL, achievable in materials such as $\text{Rb}_{1-\delta}\text{V}_2\text{Te}_2\text{O}$ and $\text{KV}_2\text{Se}_2\text{O}$ [51,52]. In this Letter, we focus on this maximal ΔP scenario and demonstrate that it generates purely spin-polarized triplet pairing in the bulk AM via proximity effect. As a proof of concept, we study a minimal d -wave AM Hamiltonian on a square lattice:

$$\mathcal{H}_{\text{AM}}(\mathbf{k}) = \epsilon_0(\mathbf{k})\hat{\sigma}_0 - 2J_{\text{AM}}(\cos k_x - \cos k_y)\hat{\sigma}_z, \quad (1)$$

where $\epsilon_0(\mathbf{k}) = -2t_1(\cos k_x + \cos k_y) - 4t_2 \cos k_x \cos k_y$, $\hat{\sigma}_0$ and $\hat{\sigma}_{x/y/z}$ are identity and Pauli matrices acting on spin space, t_1 (t_2) is the nearest (next-nearest) neighbor hopping amplitude, and J_{AM} denotes the strength of the d -wave altermagnetic spin splitting. The band structure exhibits valley-dependent splittings [Fig. 2(a)]: a positive splitting $+4J_{\text{AM}}$ at X while a negative splitting $-4J_{\text{AM}}$ at Y . Thus, for a wide range of chemical potentials (around $\mu \sim 2.4$), the system hosts fully spin-polarized Fermi pocket centered at each valley [Fig. 2(b)], achieving maximal spin-valley polarization ($\Delta P = -1$). Note that this scenario is not limited to this specific model but is achievable on various lattice systems [25,70,71].

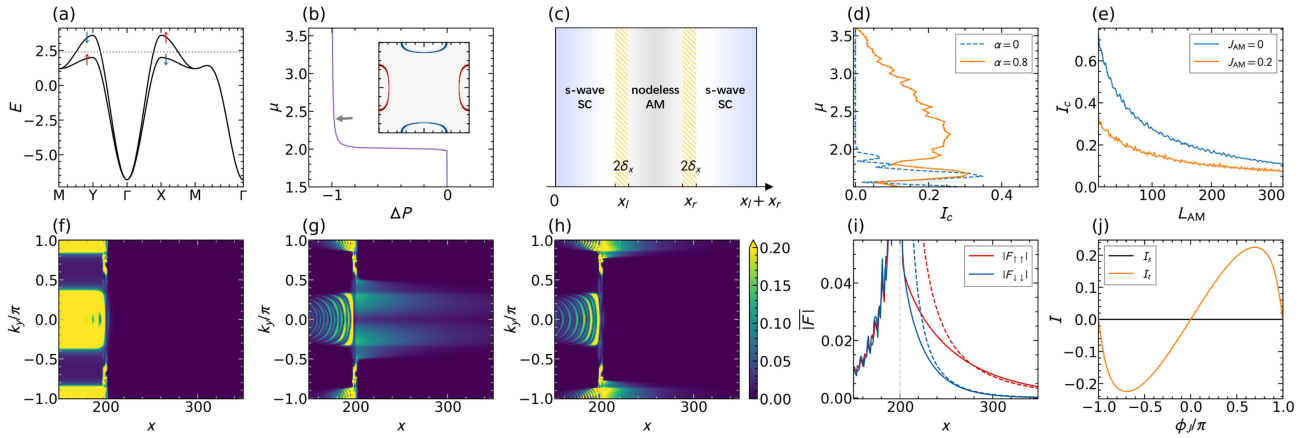


FIG. 2. Main results for 0° -aligned nodeless AM-based Josephson junctions. (a) Band structure of the AM from Eq. (1), with red ($\uparrow$) and blue ($\downarrow$) arrows denoting spin-polarized bands. (b) Spin-valley polarization ΔP as a function of μ for bands in (a). Inset: spin-split Fermi surfaces at $\mu = 2.4$ [dashed line in (a)]. (c) Schematic of the SC-AM-SC Josephson junction. The yellow region denotes the interface that breaks inversion symmetry. All results refer to this geometry ($x_l = L_{\text{SC}}$, $x_r = L_{\text{SC}} + L_{\text{AM}}$). (d) Critical Josephson current I_c in the SC-AM-SC junction with $L_{\text{AM}} = 50$ as a function of μ , comparing spin-conserving ($\alpha = 0$, blue dashed) and non-spin-conserving ($\alpha = 0.8$, orange solid) interfaces. (e) I_c versus AM length L_{AM} , with (orange) and without (blue) the AM spin-splitting term J_{AM} . (f)–(i) Proximity-induced pairing correlations near the left superconducting interface using $L_{\text{AM}} = 1600$ and frequency $\omega = 0.02$: (f) spin-singlet $|F_s(x, k_y)|$, (g) up-up triplet $|F_{\uparrow\uparrow}(x, k_y)|$, (h) down-down triplet $|F_{\downarrow\downarrow}(x, k_y)|$; and (i) spatial decay of k_y -averaged equal-spin triplet pairing amplitudes: $|F_{\downarrow\downarrow}(x)|$ (red) and $|F_{\uparrow\uparrow}(x)|$ (blue), showing distinct decay rates. (j) Current-phase relation $I(\phi_f)$ for $L_{\text{AM}} = 50$ decomposed into singlet (blue) and triplet (orange) contributions, showing exclusively triplet-driven supercurrent. Parameters: $t_1 = 1$, $t_2 = 0.7$, $\mu = 2.4$, $J_{\text{AM}} = 0.2$, $\Delta_0 = 0.02$, $\alpha = 0.8$, $T = 0.02\Delta_0$, and $L_{\text{SC}} = 200$.

Josephson junctions based on nodeless AMs—We next explore the role of maximal spin-valley polarization in SC-AM-SC Josephson junctions [Fig. 2(c)]. We consider a planar junction formed by two s -wave superconductors (SCs) separated by a nodeless AM. For junctions oriented along the x -direction, the two valleys in the AM become fully decoupled, and each behaves like a half metal. This effectively creates two parallel half-metallic transport channels that naturally carry spin-polarized triplet Josephson currents. To analyze the Josephson effect, we model the junction with the Hamiltonian

$$\mathcal{H}_{\text{SNS}} = \sum_{k_y} (\mathcal{H}_0 + \mathcal{H}_{\text{L1}} + \mathcal{H}_{\text{AM}} + \mathcal{H}_{\text{L2}} + \mathcal{H}_{\text{SC-AM}}), \quad (2)$$

in Nambu basis $C_x^\dagger = (c_{x\uparrow}^\dagger, c_{x\downarrow}^\dagger, c_{x\uparrow}, c_{x\downarrow})$. We assume translation symmetry along the interface, so k_y remains a good quantum number. The kinetic term, $\mathcal{H}_0 = \sum_x \{C_x^\dagger (-2t_1 \cos k_y - \mu) \hat{\tau}_z \hat{\sigma}_0 C_x + [C_{x+1}^\dagger (-t_1 - 2t_2 \cos k_y) \times \hat{\tau}_z \hat{\sigma}_0 C_x + \text{H.c.}]\}$, acts throughout the entire system. $\hat{\tau}_{x/y/z}$ are Pauli matrices in particle-hole space. The on-site s -wave pairing terms in the two SC leads are $\mathcal{H}_{\text{L1}} = -\Delta_0 \sum_{0 \leq x < x_l} C_x^\dagger \hat{\tau}_y \hat{\sigma}_y C_x$ with $x_l = L_{\text{SC}}$ and $\mathcal{H}_{\text{L2}} = -\Delta_0 \sum_{x_r \leq x < x_r + L_{\text{SC}}} C_x^\dagger [\cos \phi_J \hat{\tau}_y - \sin \phi_J \hat{\tau}_x] \hat{\sigma}_y C_x$ with $x_r = x_l + L_{\text{AM}}$, where Δ_0 is the pairing gap and ϕ_J is the superconducting phase difference. The term for nodeless AM in the junction reads $\mathcal{H}_{\text{AM}} = J_{\text{AM}} \sum_{x_l \leq x < x_r} [C_x^\dagger (2 \cos k_y) \times \hat{\tau}_z \hat{\sigma}_z C_x - (C_{x+1}^\dagger \hat{\tau}_z \hat{\sigma}_z C_x + \text{H.c.})]$. The interfacial Rashba spin-orbit coupling, arising from structural inversion symmetry breaking [72], is confined to a finite interfacial region, i.e., $\mathcal{H}_{\text{SC-AM}} = \alpha \sum_{|x-x_{l/r}| \leq \delta_x} \{C_x^\dagger (-\sin k_y) \hat{\tau}_0 \hat{\sigma}_x C_x + [C_{x+1}^\dagger (i/2) \hat{\tau}_z \hat{\sigma}_y C_x + \text{H.c.}]\}$. It can be achieved in lateral junctions with SC leads on the AM boundaries or via local electrostatic gating [73]. While we use $\delta_x = 2$, our main conclusions remain unaffected by this choice.

Based on the continuity equation [see Sec. S1 in Supplemental Material (SM) [74]], we calculate the local supercurrent flowing across the junction as [75–77]

$$I_x(\phi_J) = -\frac{4e}{\hbar\beta} \sum_{\omega, k_y} \text{Im} \left[\text{Tr} [\hat{T}_h^\dagger F_{x+1} \hat{T}_e \tilde{F}_x] \right], \quad (3)$$

where $\beta = 1/k_B T$, $\omega = (2n+1)\pi/\beta$ are Matsubara frequencies, $\hat{T}_{e/h}(k_y) = \mp [(t_1 + 2t_2 \cos k_y) \hat{\sigma}_0 + J_{\text{AM}} \hat{\sigma}_z]$ are electron (hole) hopping matrices, respectively, $F_x(\omega, k_y)$ is the anomalous Green's function at site x inside the AM, and $\tilde{F}_x(\omega, k_y)$ is the surface anomalous Green's function [78]. Details are provided in Sec. S1 of SM [74]. The current is uniformity within the AM, i.e., $I_x = I$ throughout the junction. The critical current I_c is defined as the extreme value of $I(\phi_J)$ within $0 < \phi_J < \pi$.

Figure 2(d) presents I_c as a function of chemical potential μ . We set $\Delta_0 = 0.02t_1$, corresponding to a coherence length $\xi_{\text{SC}} \approx 160$. For $\alpha = 0$, I_c vanishes even for a short junction with $L_{\text{AM}} \geq 16$ ($\sim 0.1\xi_{\text{SC}}$) for μ in the maximal spin-valley polarization region. This suppression stems from spin $U(1)$ symmetry, which restricts the junction to spin-singlet pairing correlations that decay rapidly in the nodeless AM. We will substantiate this point explicitly later. However, introducing a finite α -term breaks spin-rotation symmetry at the interfaces, enabling singlet-to-triplet conversion and leading to finite I_c [solid orange line for $\alpha = 0.8$, Fig. 2(d)]. Alternatively, such conversion can be achieved using spin-orbit-coupled s -wave SCs or interfacial spin-canting [see Sec. S2 in SM [74]]. Remarkably, at $\alpha = 0.8$, a significant I_c emerges for both short and long junctions (e.g., $L_{\text{AM}} = 300 \sim 2\xi_{\text{SC}}$), with magnitude comparable to the nonmagnetic counterpart ($J_{\text{AM}} = 0$) [Fig. 2(e)].

Proximity-induced pure triplet correlations—Equation (3) demonstrates that the on-site pairing correlations fully govern the supercurrent. To understand the microscopic origin of the pronounced I_c , we analyze the proximity-induced pairing correlations within the nodeless AM, extracted from $F_x(\omega, k_y)$ in Eq. (3). These correlations decompose as

$$F = -i\hat{\sigma}_y F_s + \frac{\hat{\sigma}_0 + \hat{\sigma}_z}{2} F_{\uparrow\uparrow} + \frac{\hat{\sigma}_0 - \hat{\sigma}_z}{2} F_{\downarrow\downarrow} + \hat{\sigma}_x F_z, \quad (4)$$

with $F_{s/z} = (F_{\downarrow\uparrow} \mp F_{\uparrow\downarrow})/2$. F_s corresponds to the singlet pairing, while F_z , $F_{\uparrow\uparrow}$, and $F_{\downarrow\downarrow}$ represent the triplet pairings. To clarify the behavior of induced pairings, we consider a long SC-AM-SC junction with $L_{\text{AM}} = 1600 \approx 10\xi_{\text{SC}}$ to minimize finite-size effects. As shown in Fig. 2, only spin-triplet correlations exhibit long-range proximity effects in the AM. Here, only results near the left SC interface are shown. Specifically, the singlet component $|F_s|$ decays rapidly for all k_y [Fig. 2(f)]. In sharp contrast, the equal-spin triplet components $|F_{\uparrow\uparrow}|$ and $|F_{\downarrow\downarrow}|$, which arise at finite frequencies, penetrate deeply into the AM [Figs. 2(g) and 2(h)]. This defines the pure triplet correlations in AM: $|F_s| = 0$ while $|F_{\uparrow\uparrow}| \neq 0$ and/or $|F_{\downarrow\downarrow}| \neq 0$. Similar behavior is demonstrated for a short junction ($L_{\text{AM}} = 50$) in Sec. S2 of SM [74]. These proximity-induced triplet pairings exhibit the same symmetry-breaking characteristics as the bulk AM and thus can be classified as extrinsic altermagnetic SCs [69]. As a result, they are fundamentally distinct from the p -wave triplet states that arise from altermagnetic fluctuations [79].

The proximity-induced triplet pairing is intrinsically spin-polarized, manifested in distinct decay rates of the k_y -averaged $|F_{\uparrow\uparrow}|$ and $|F_{\downarrow\downarrow}|$ [Fig. 2(i)]. This polarization originates from valley-locked pairing correlations: $|F_{\uparrow\uparrow}|$ emerges exclusively from the X -valley Fermi surface with effective Fermi velocity $v_{F,\uparrow} \approx 3.3$ along the transport

direction, while $|F_{\downarrow\downarrow}|$ stems solely from the Y -valley with $v_{F,\downarrow} \approx 1.2$. These velocities qualitatively determine the decay lengths λ_σ (via $\lambda_\sigma \propto v_{F,\sigma}$) that yield fits to the decay profiles using $|F_{\sigma\sigma}| \propto (1/x)e^{-x/\lambda_\sigma}$ for clean systems, directly governing the observed decay asymmetry in Fig. 2(i). We demonstrate that the valley-spin locked pairing directly encodes maximal spin-valley polarization ($\Delta P = -1$), and confirm vanishing finite-size magnetization for large $L_{AM} \gg 1/k_F$ [80,81]. Thus, these pairing correlations naturally drive the magnetization-free spin-polarized Josephson supercurrent. Since the hopping matrices $\hat{T}_{e/h}$ in Eq. (3) are diagonal in spin space, the supercurrent $I(\phi_J)$ decomposes as

$$I(\phi_J) = I_s(\phi_J) + I_t(\phi_J) + I_{st}(\phi_J), \quad (5)$$

where $I_s \propto F_s \tilde{F}_s$ and $I_t \propto F_{\uparrow\uparrow} \tilde{F}_{\uparrow\uparrow} + F_{\downarrow\downarrow} \tilde{F}_{\downarrow\downarrow} + F_z \tilde{F}_z$ [see Sec. S1 of SM [74]]. Due to the pure-triplet nature of the induced pairing in AM (i.e., $F_s = \tilde{F}_s = 0$), the singlet-triplet mixing contribution (I_{st}) to I vanishes. As shown in Fig. 2(j), I_s is zero for all ϕ_J , leaving triplet correlations as the sole source of supercurrent. The triplet supercurrent polarization ratio is $I_{t,\uparrow\uparrow}/I_{t,\downarrow\downarrow} \approx 3.4$ for our parameters [see Sec. S3 in SM [74]]. Notably, this polarization is essentially governed by $v_{F,\uparrow}/v_{F,\downarrow}$ and can be enhanced by tuning this ratio. Our results demonstrate a spin-polarized supercurrent in alternating magnetic SCs. This feature is generic and can also be realized in Rashba SC-nodeless AM-Rashba SC junctions [see Sec. S3 of SM [74]]. Crucially, unlike ferromagnet-based junctions, our results for AM-based junction does not require any net magnetization [82].

Effects of junction orientation—The robustness of these results originates from the forbidden intervalley spin-singlet pairing channel. We now show how junction orientation controls the emergence of this channel. In a 45° -aligned junction [Fig. 3(a); see Sec. S2 in SM [74]], the global Fermi surface rotation hybridizes the X and Y valley indices in the rotated k_1 - k_2 frame. While k_2 remains conserved, this mixing enables both intravalley (e.g., $\langle c_{X,\uparrow} c_{X,\uparrow} \rangle$) and intervalley (e.g., $\langle c_{X,\uparrow} c_{Y,\downarrow} \rangle$) pairings via proximity. In Fig. 3(b), we find both $I_s(\phi_J)$ and $I_t(\phi_J)$ contribute to the supercurrent, while the spin-singlet correlation persists even at $\alpha = 0$ [see Sec. S4 in SM [74]]. For $\alpha \neq 0$, Figs. 3(c) and 3(d) show coexisting singlet and triplet correlations throughout the nodeless AM. Thus, rotating the junction orientation from 0° to 45° induces a crossover from pure triplet to mixed singlet-triplet supercurrent, although triplet pairing remains dominant. This orientation dependence directly manifests the anisotropic spin-splitting inherent to AM, fundamentally distinguishing AM-based junctions from other systems such as half metals [83].

Tunable $0-\pi$ transition—The Josephson current mediated by triplet pairings can be controlled to realize tunable $0-\pi$ transition. To illustrate this, we first analyze how the

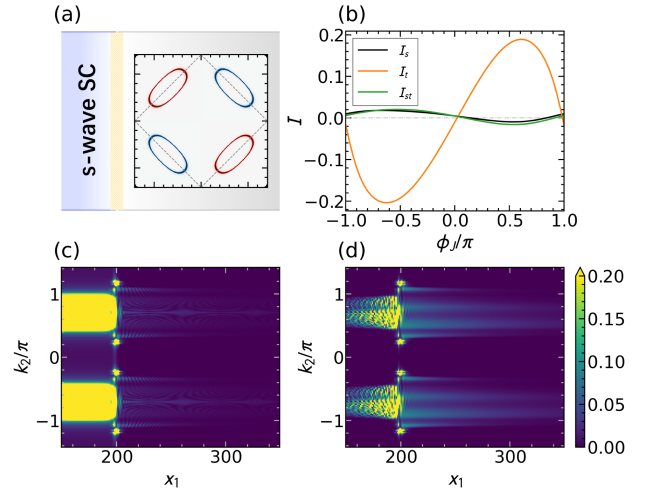


FIG. 3. (a) Sketch of the SC-AM junction. Inset: Fermi surface of the AM rotated by $\pi/4$, with gray dashed line marking the first Brillouin zone boundary. (b) Phase-dependent Josephson currents: singlet (I_s), triplet (I_t), and mixed (I_{st}) contributions versus phase difference ϕ_J in the SC-AM-SC junction. Spatial evolution of $|F_s(k_2)|$ in (c) and $|F_{\uparrow\uparrow}(k_2)|$ in (d) near the SC-AM interface.

interfacial spin-orbit coupling affects the pairing correlations in the NS junction. To incorporate valley degrees of freedom, we compute the k_y -summed pairing correlations, $F_i(\omega) = 1/2\pi \int_{-\pi}^{\pi} dk_y F_i(\omega, k_y)$, where F_i (with $i \in \{s, z, \uparrow\uparrow, \downarrow\downarrow\}$) are defined in Eq. (4).

For the 0° junction, Fig. 4(a) shows F_s , $F_{\uparrow\uparrow}$, and $F_{\downarrow\downarrow}$ as functions of α at $x = L_{SC} + 20$, deep within the AM bulk.

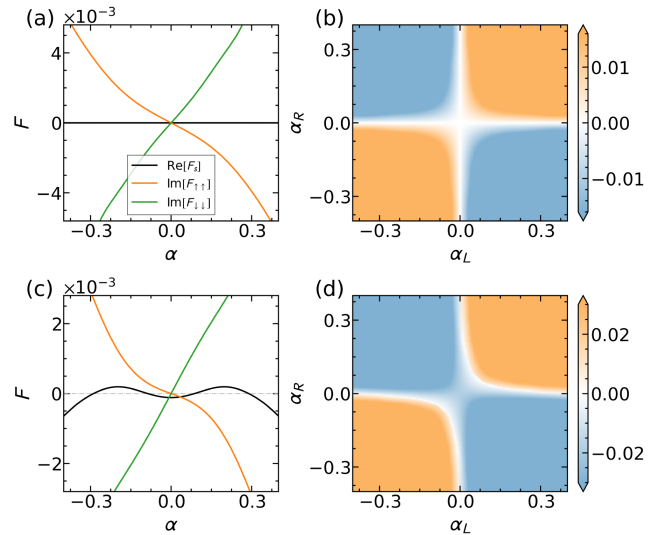


FIG. 4. (a) Dependence of pairing correlations on interfacial Rashba strength α in the 0° junction. (b) Critical current phase diagram: I_c versus interfacial Rashba couplings α_L and α_R in the SC-AM-SC junction. (c) Dependence of pairing correlations on interfacial Rashba strength α in the 45° -junction. (d) Critical current $I_c(\alpha_L, \alpha_R)$ exhibiting a butterfly pattern in the 45° -junction.

While F_s vanishes at this distance ($20 > 0.1\xi_{\text{SC}}$), both triplet components $F_{\uparrow\uparrow}$ and $F_{\downarrow\downarrow}$ emerge and grow with $|\alpha|$. Notably, the triplet components reverse sign when α changes sign, $\alpha \rightarrow -\alpha$, indicating their sensitivity to the sign of the interfacial spin-orbit coupling. Extending to the SNS junction, we introduce two independent interfacial spin-orbit coupling strengths α_L and α_R at the left and right interfaces, respectively. Remarkably, by tuning these two parameters, we observe robust $0-\pi$ transitions in the Josephson current [Fig. 4(b)]. The phase boundaries lie along the $\alpha_L = 0$ and $\alpha_R = 0$, intersecting to form a cross (+) symbol. Crucially, it follows $\text{sign}[I_c] = \text{sign}[\alpha_L \alpha_R]$, demonstrating that the relative sign of spin-orbit coupling at the interfaces determines whether the junction is in the 0 or π state.

In the 45° junction, F_s coexists with $F_{\uparrow\uparrow}$ and $F_{\downarrow\downarrow}$ [Fig. 4(c)]. Since F_s is an even function of α , however, the sharp $0-\pi$ transition lines at $\alpha_L = \alpha_R = 0$ become avoided crossings. Consequently, this transforms the $0-\pi$ boundaries in the α_L - α_R plane into a distinctive butterfly pattern [Fig. 4(d)].

Discussions and conclusions—Finally, we note that the spin-triplet Josephson current I_t contains contributions from both even- ω and odd- ω triplet correlations, as shown in the ω -summation in Eq. (3). Explicitly, the triplet pairing can be decomposed as $F_{\uparrow\uparrow(\downarrow\downarrow)} = F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{even}} + F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{odd}}$ with $F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{even}}(\omega) = F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{even}}(-\omega)$ and $F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{odd}}(\omega) = -F_{\uparrow\uparrow(\downarrow\downarrow)}^{\text{odd}}(-\omega)$ [84]. Accordingly, I_t in Eq. (5) can be separated into two parts,

$$I_t(\phi_J) = I_t^o(\phi_J) + I_t^e(\phi_J), \quad (6)$$

with $I_t^{e(o)} \propto F_{\uparrow\uparrow}^{\text{even(odd)}} \tilde{F}_{\uparrow\uparrow}^{\text{even(odd)}} + F_{\downarrow\downarrow}^{\text{even(odd)}} \tilde{F}_{\downarrow\downarrow}^{\text{even(odd)}}$. Hence, I_t^o can serve as a direct detector for odd- ω triplet pairing when $I_s(\phi_J) = I_t^e(\phi_J) = 0$. As shown in Sec. S3 of SM [74], the interfacial spin-orbit coupling induces coexisting even- ω and odd- ω triplets, yielding finite I_t^o and I_t^e . In contrast, an interfacial spin canting, described by $\mathcal{H}_{\text{SC-AM}} \propto \sum_{|x-x_{I/r}| \leq \delta_x} C_x^\dagger [\cos(x\pi/2) \hat{\tau}_z \hat{\sigma}_x + \sin(x\pi/2) \hat{\tau}_0 \hat{\sigma}_y] C_x$, produces purely odd- ω triplets, resulting in only I_t^o . This provides a direct signature of odd- ω spin-triplet pairing in Josephson current measurements.

In summary, we have shown that nodeless altermagnets with maximal spin-valley polarization provide a unique platform for generating pure spin-triplet Josephson currents without net magnetization. The valley-locked pairing mechanism, in which two equal-spin triplet pairing correlations originate exclusively from two separated valleys, respectively, enables long-range triplet proximity effects unattainable in conventional metals or nodal altermagnets. Crucially, this system exhibits two experimentally tunable control knobs: (i) junction orientation, which governs the triplet purity and enables a crossover from exclusive triplet supercurrents (0° junction) to hybrid singlet-triplet states (45° junction); and (ii) interfacial symmetry breaking

(α_L and α_R), which triggers robust $0-\pi$ transitions without fine tuning, following the sign rule $\text{sign}[I_c] = \text{sign}[\alpha_L \alpha_R]$.

Experimental realization of our proposal is feasible using well-established fabrication techniques with spin-valley-locked altermagnets, such as $\text{KV}_2\text{Se}_2\text{O}$ [51], $\text{Rb}_{1-\delta}\text{V}_2\text{Te}_2\text{O}$ [52], and $\text{SrFe}_4\text{O}_{11}$ [85]. Notably, the predicted spin-triplet Josephson supercurrent exhibits exceptional robustness against Zeeman fields [see Sec. S5 in SM [74]], which provides a distinctive signature contrasting sharply with singlet-dominant supercurrent. Our findings thereby establish nodeless altermagnets as a functional material platform for *magnetization-free superconducting spintronics*. Combining our results with prior studies [57–66] yields a comprehensive framework for superconducting proximity effects in altermagnets, thereby establishing the theoretical basis for exotic altermagnetic superconductors [69].

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Data availability—The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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